

CUSTOMER

PROJECT	x				
MANUFACTURER					
DOCUMENT NO.	x				
DOC TITLE	Ampacity Calculation				
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	Unit	Reference	Clause	Symbol	Formula
					TREFOIL GNDTREFOIL DUCTTREFOIL AIR

DESIGN DATA

Bonding Type	Design data		solid	solid	solid
Cond Mat	Design data		Al	Al	Al
Cond size	mm²	Design data	185	185	185
Cond design	Design data		CC	CC	CC
Cond Dn	mm	Design data	Dc	15.9	15.9
Cond Sc Dn	mm	Design data	Ds		
Ins Mat	Design data		XLPE	XLPE	XLPE
Ins Dn	mm	Design data	Di	33.9	33.9
Ins Sc Dn	mm	Design data	dx		
Ins Tn	mm	Design data	Ti	9.19	9.19
Core Dn	mm	Design data	Do		
Lay-up OD	mm	Design data	Da		
Cable Dn	mm	Design data	De	43.4	43.4
Core No	nos	Design data	n		
Wire No	nos	Design data	nw		
Cable weight	kg/km	Design data	M		
Max cond temp	°C	Design Data	θmax	90	90
Earth fault current	A	Design Data	I■/1sec	1493	1493
Rated voltage	kV	IEC60502-2 Table 1	U		
Earth voltage	kV	IEC60502-2 Table 1	U■	19.053	19.053
Impulse voltage	kV	IEC60502-2 Table 14	Up		

PARAMETER SETTING

Soil TR	K.m/W	Input	ρ	1.2	1.2	1.2
Multicore factor		Input	fm	1	1	1
Frequency	Hz	Input	f	50	50	50
Ambient temp.	°C	Input	θa	30	30	40
Depth	mm	Input	L	1000	1000	1000
Duct Type		Input		HDPE	HDPE	HDPE
External diameter of duct	mm	Input	dc	110.0	110.0	110.0
Thickness of duct	mm	Input	Td	0	5.0	5.0
Duct temperature	°C	Input	θm	85	85	85
Temp. rise, ambient to max.	°C	IEC60287-1-1 1.4.1.1	Δθ	θmax - θa	60	50
Core temperature, θc	°C	Iteration	θc	80	78	75

DC RESISTANCE

DC resistant at 20°C	Ω/m	IEC60228 Class 2	R■	1.640E-04	1.640E-04	1.640E-04
DC resistant at ambient	Ω/m	IEC60287-1-1 2.1.1	R■ amb	R■(1 + α■(θx - 20))	1.706E-04	1.772E-04
DC resistant at 90°C	Ω/m	IEC60287-1-1 2.1.1	R■ max	R■(1 + α■(θmax - 20))	2.103E-04	2.103E-04

AC RESISTANCE

Strand constant		IEC60287-1-1 Table 2	ks	1.0	1.0	1.0
Strand constant		IEC60287-1-1 Table 2	kp	0.8	0.8	0.8
Spacing correction		Mc Graw Hill Page 161		1.250	1.250	1.250
Bessel function		IEC60287-1-1 2.1.2	Xs■	[ks (8πf/Ro) 10■]²	0.587	0.587
Skin effect		IEC60287-1-1 2.1.2	Ys	Xs■/(192 + 0.8 Xs■)	0.003	0.003
Proximity function		IEC60287-1-1 2.1.4	Xp■	[kp (8πf/Ro) 10■]²	0.376	0.376
Proximity function		IEC60287-1-1 2.1.4	Yp"	Xp■/(192 + 0.8 Xp■)	0.002	0.002
Proximity effect		IEC60287-1-1 2.1.4	Yp	Yp" (Dc/s)² [0.312 (Dc/s)² + 1.18/(Yp" + 0.27)]	0.001	0.001
AC resistant at 20°C	Ω/m	IEC60287-1-1 2.1	R■	R■ (1 + Ys + Yp)	1.647E-04	1.647E-04
AC resistant at 20°C - isolated	Ω/m	IEC60287-1-1 2.1	R'■	R■ (1 + Ys)	1.645E-04	1.645E-04
AC resistant at 90°C	Ω/m	IEC60287-1-1 2.1	R■	R■ max (1 + Ys + Yp)	2.108E-04	2.107E-04

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AC resistant at 90°C - isolated	Ω/m	IEC60287-1-1 2.1		R'■■■	R■■ max (1 + Ys)	2.107E-04	2.107E-04	2.107E-04
SHEATH LOSS FACTOR								
Equivalent mean diameter	mm	Design data		ds		38.0	38.0	38.0
Equivalent resistant at 20°C	Ω/m	IEC60287-1-1		R■■■	ρ/A	3.448E-04	3.448E-04	3.448E-04
Equivalent resistant at amb	Ω/m	IEC60287-1-1		Rx	R■■■(1 + α■■■(θx - 20)	7.180E-04	7.180E-04	7.450E-04
Equivalent resistant at θc	Ω/m	IEC60287-1-1		Rs	R■■■(1 + α■■■(θc - 20)	8.550E-04	8.490E-04	8.400E-04
Sheath Inductance	H/m	IEC60287-1-1 2.3.3		L	2ln(2s/d) x 10■■■	1.680E-07	4.260E-07	1.680E-07
Sheath Reactance	Ω/m	IEC60287-1-1 2.3.3		X	2ωln(2s/d) x 10■■■	5.290E-05	1.340E-04	5.290E-05
Circulating current loss factor		IEC60287-1-1 2.3.1		λ■■'	Rs/R /(1 + (Rs/X)²	0.033	0.134	0.033
Eddy current loss factor		IEC60287-1-1 2.3.6.1		λ■■''		0.000	0.000	0.000
Sheath loss factor		IEC60287-1-1 2.3		λ■■	λ■■' + λ■■''	0.033	0.134	0.033
THERMAL RESISTANCE								
T1 factor		IEC60287-2-1				1.000	1.000	1.000
Thermal Resistant Insulation	K.m/W	IEC60287-2-1		T■■	(ρ/2π)LN[1 + (2t/dc)]	0.451	0.451	0.451
T2 factor		IEC60287-2-1				1.000	1.000	1.000
Thermal Resistant filler	K.m/W	IEC60287-2-1		T■■f	0.031(pf - pi)exp(0.67ti/dc))	0.000	0.000	0.000
Thermal Resistant Bedding	K.m/W	IEC60287-2-1		T■■	(ρ/2π)LN[1 + (2t/ds)]	0.000	0.000	0.000
T3 factor		IEC60287-2-1				1.600	1.600	1.000
Thermal Resistant Outercovering	K.m/W	IEC60287-2-1		T■■	(ρ/2π)LN[1 + (2t/D)]	0.208	0.130	0.130
Thermal Resistant External	K.m/W	IEC60287-2-1 2.2		T■■		2.231	2.377	1.011
CURRENT RATING								
Critical Temperature Rise of Soil	°C	IEC60287-1-1		dθ	Wk (ρc /2π)LN(Gf)	0.0	0.0	0.0
Top Equation		IEC60287-1-1 1.4.1		Etop	Δθ - Wd(0.5T■■ + n(T■■ + T■■) + T■■) - dθ	59.9	59.9	50.0
Bottom Equation		IEC60287-1-1 1.4.1		Ebot	RT■■ + nR(1 + λ■■)T■■ + nR(1 + λ■■ + λ■■)(T■■,267■■)-04	2.67E-04	6.939E-04	3.438E-04
Calculated Amps	A	IEC60287-1-1 1.4.1		I	√(Etop/Ebot)	309	294	381
Calculated MVA	MVA	IEC60287-1-1 1.4.1		MVA	√3UI	18	17	22
ELECTRICAL DATA								
Inductance	μH/m	IEC60287-1-1 2.3		L	0.2LN(2s/Dm)	0.168	0.426	0.168
Reactance	μΩ/m	IEC60287-1-1 2.3		X	ω Lm x 10■■³	52.9	133.9	52.9
Pos Seq Resistance	μΩ/m			R■■	R'■■ (1 + λ■■ + λ■■)	214.9	236.4	214.9
Pos Seq Inductance	μH/m			L■■	fw+0.2 LN(2s/dc)	0.395	0.653	0.395
Pos Seq Reactance	μΩ/m			X■■	ω L x 10■■³	124.1	205.2	124.1
Pos Seq Impedance	μΩ/m			Z■■	√(R■■²+Xm²)	244.6	294.1	244.6
Zero Seq Resistance	μΩ/m			R■■	R■■ + (n*R'■■)	889.2	889.2	923.0
Zero Seq Inductance	μH/m			L■■	fw+0.2LN(Dm/dc)	0.227	0.227	0.227
Zero Seq Reactance	μΩ/m	BICC Chap 10		X■■	ω L x 10■■³	71.3	71.3	71.3
Zero Seq Impedance	μΩ/m	BICC Chap 10		Z■■	√(R■■²+X■■²)	892.1	892.1	925.7
Capacitance	μF/km	IEC60287-1-1 2.2		C	ε/(18LN(Di/ds)	2.119E-10	2.119E-10	2.119E-10
Charging current - Max	A/km	BICC Page 170		Ic max	1.08 ωCU■■	1.25	1.25	1.25
Surge velocity		Tanaka Vol.1 Page 179		Vs		170	170	170
Surge inductance		Tanaka Vol.1 Page 179		Ls	1/CVs²	1.785E-04	1.785E-04	1.785E-04
THERMAL LOSS								
Thermal Resistant Ins	K.m/W	IEC60287-2-1		T■■	(ρ/2π)LN[1+(2t/dc)]	0.451	0.451	0.451
Thermal Resistant Ish	K.m/W	IEC60287-2-1		T■■	T■■ + T■■f	0.000	0.000	0.000
Thermal Resistant Osh	K.m/W	IEC60287-2-1		T■■	(ρ/2π)LN[1+(2t/D)]	0.208	0.130	0.130
Thermal Resistant Ext	K.m/W	IEC60287-2-1 2.2		T■■		2.231	2.377	1.011
Thermal Resistant Total	K.m/W	Mc Graw hill Page 34		ΣT	T■■ + T■■ + T■■ + T■■	2.969	2.446	1.581
Temp Over Ins	°C	Mc Graw hill Page 34		θ■■	θ-W*T■■	80.32	78.25	74.85
Temp Over Ish	°C	Mc Graw hill Page 34		θ■■	θ■■-W*T■■	80.32	78.25	74.85
Temp Over Osh	°C	Mc Graw hill Page 34		θ■■	θ■■-W*T■■	77.03	74.25	71.63

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Temp Ambient	°C	Mc Graw hill Page 34		θ_i	$\theta_i - W \cdot T$	30.00	30.00	40.00
Thermal loss	W/m	Mc Graw hill Page 34		W	$\theta_c - \theta_a / \Sigma T$	20.21	24.53	31.62
Thermal loss Ins	W/m			W _{ins}		3.26	4.81	9.58
Thermal loss Ish	W/m			W _{sh}		0.00	0.00	0.00
Thermal loss Osh	W/m			W _{osh}		1.11	1.63	2.04
Thermal loss Ext	W/m			W _{ext}		15.84	18.09	20.00
Losses - Conductor	W/m	IEC60287-1-1 2.4.2		W _c	$3(I^2R)$	20.18	18.20	30.64
Losses - Dielectric	W/m	IEC60287-1-1 2.2		W _d	$3\omega CUo^2\tan\delta$	0.00	0.00	0.00
Losses - Metal sheath	W/m	IEC60287-1-1	W _m	W _{cλ}		1.15	8.19	1.84
INDUCED PARAMETER								
Normal - Sheath voltage	V/m	IEEE Std 575 Page 29		U _s	IX	0.016	0.044	0.020
Normal - Circulating current	A	IEEE Std 575 Page 29		I _s	$\sqrt{(I^2(R/R_s)\lambda_{ins})}$	21.20	56.71	27.01
Transient - Sheath voltage	V/m	IEEE Std 575 Page 29		U _{st}	I _{st} X	0.079	0.200	0.079
Transient - Circulating current	A	IEEE Std 575 Page 29		I _{st}	$\sqrt{(I_{st}^2(R/R_s)\lambda_{ins})}$	103	260	105
VOLTAGE DROP								
Rated voltage	kV	IEC60840 Table 4		U		33	33	33
Pos Seq Impedance	μΩ/m			Z _{seq}	$\sqrt{(R_{seq}^2+X_{m^2})}$	244.6	294.1	244.6
Calculated Amps	A	IEC60287-1-1 1.4.1		I	$\sqrt{(E_{top}/E_{bot})}$	309	294	381
Permissible limit	%	Input		a		2.5%	2.5%	2.5%
Permissible voltage drop	V			E	a*U	825	825	825
Voltage drop factor	mV/A/m			R	$\sqrt{3}*Z$	0.424	0.509	0.424
Maximum permissible length	km			L	$E/(I*R)$	6.3	5.0	5.1
CRITICAL LENGTH								
Frequency	Hz	Input		f		50	50	50
Capacitance	μF/km	IEC60287-1-1 2.2		C	$\epsilon/(18LN(D_i/d_s))$	2.119E-10	2.119E-10	2.119E-10
Calculated Amps	A	IEC60287-1-1 1.4.1		I	$\sqrt{(E_{top}/E_{bot})}$	309	294	381
Rated Voltage	kV	IEC60840 Table 4		U		33	33	33
Critical Length	km			L _{max}	$\sqrt{3(I/\omega CU)} \times 10^3$	265	281	332
RECEIVED POWER								
Frequency	Hz	Input		f		50	50	50
Length	km	Input		L		30	30	30
Max Feeding Power	MVA	IEC60287-1-1 1.4.1		S		17.6	18.6	22.0
Rated Voltage	kV	IEC60840 Table 4		U		33	33	33
Capacitance	μF/km	IEC60287-1-1 2.2		C	$\epsilon/(18LN(D_i/d_s))$	2.119E-10	2.119E-10	2.119E-10
Cable Reactive Power	MVAR			Q _{cable}	$\omega CL U^2 \times 10^{-3}$	2.0	2.0	2.0
Max Received Power	MW			P	$\sqrt{[S^2 - Q^2]}$	17.4	18.5	21.9
Power Losses	%			P _{loss}	P/S	0.6%	0.6%	0.4%
